

Ray Sarabu

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EDUCATION

Carnegie Mellon University

B.S. Electrical and Computer Engineering + Robotics; GPA: 3.65/4.0

Pittsburgh, PA

Aug 2024 – May 2027 (Expected)

- **Relevant coursework:** General Robotics, Fundamentals of Control, Digital System Design, Embedded Systems, Analog Circuits, Signals and Systems, Computer Systems
- **Teaching Assistant**, Reasoning with Data (36-200): led weekly labs and office hours for 40+ students covering statistical reasoning, R programming, and data analysis

PROFESSIONAL EXPERIENCE

General Motors — Advanced Build and Integration Team

Advanced Systems Development Intern

Warren, MI

May – Aug 2026

- Helped build GM's ADAS vehicle-checkout tool for automated setup and verification, implementing 35 workflow steps and 76 pass/fail checks across 5 LiDARs, 8 radars, and 11 cameras to replace a manual pre-run checklist.
- Built a visualization tool that parses 128-channel LiDAR UDP multicast streams in real time and reconstructs full 3D point clouds directly from raw MSOP sensor packets.
- Applied camera-LiDAR sensor fusion to project point clouds onto camera fields of view, giving vehicle operators immediate visual confirmation of sensor alignment and data quality before collection runs.
- Designed and deployed an internal React/Node operations dashboard that polls live build data every 30s, giving build teams real-time visibility into blockers, per-vehicle build status, and issue ownership.

CMU MetaMobility Lab — Wearable Robotic Knee Exoskeleton

Student Researcher

Pittsburgh, PA

May 2025 – Present

- Prototyped 4 functional generations of the lab's first robotic knee exoskeleton, improving embedded electronics, mechanical integration, and hardware design to achieve 20 N-m of torque transfer with better wearability.
- Engineered the embedded control stack on Teensy and ESP32 microcontrollers, integrating IMU sensing and SPI/I2C/UART/CAN communication for real-time, low-latency robotic control.
- Implemented over-the-air (OTA) firmware updates, enabling rapid controller iteration during human-subject testing without disassembling the device.
- Developed impedance-based controllers and spline trajectory generation to align exoskeleton torque assistance with the natural walking gait cycle.

ACE Metal Crafts — Custom Stainless Steel Manufacturer

Project Engineer Intern

Chicago, IL

May – Aug 2024

- Delivered precision stainless steel components for industrial clients alongside machinists on the production floor.
- Applied machining techniques and value engineering principles to reduce fabrication cost while meeting strict dimensional and quality standards.

PROJECTS

Self-Balancing Robot | IMU Odometry, PID Control

Jan – May 2026

- Designed and tuned a self-balancing differential-drive robot using IMU odometry and PID control, tuning gains to reject disturbances and maintain balance under push perturbations.
- Fused accelerometer and gyroscope readings with a two-state Kalman filter for drift-corrected tilt estimation at a 100 Hz control loop.

Search-and-Rescue Robot | Computer Vision, Autonomous Navigation

Jan – May 2026

- Designed and built a rugged urban search-and-rescue (USAR) robot for disaster-response environments, scaling stairs, climbing 45° inclines, and traversing debris-strewn terrain while surviving 3 ft impact drops.
- Engineered a full perception-and-autonomy stack spanning OpenCV computer vision, line following, and odometry-based localization that enabled autonomous course navigation and obstacle clearing to reach simulated victims.

Robotic Arm | Inverse Kinematics, Path Planning

Aug – Dec 2025

- Developed control pipelines for a robotic arm, using inverse kinematics and path planning for end-effector positioning within 1 cm of target poses.

Electric Vehicle Builds | Enclosed Recumbent Trike, Drift Cart

May 2023 – Aug 2024

- Built two personal electric vehicles from concept to completion, integrating 48V battery packs, motor controllers, and throttle/brake systems reaching 30 mph with custom structural components via CAD, wiring, and assembly.

SKILLS

Programming: Python, C, C++, Java, JavaScript, R

Embedded & Robotics: ESP32, Teensy, Raspberry Pi, SPI/I2C/UART/CAN, PCB design, IMU sensing, PID/impedance control

Tools & Frameworks: OpenCV, NumPy, React, Node.js, Linux, Git, KiCad, SolidWorks, Fusion 360, Inventor

Activities: Choreography Chair (Greek Sing), CMU Men's Volleyball, Phi Delta Theta, weightlifting, cooking, pickleball